

Master GEO & SEO Quality Control (QC) Matrix

1. Technical Data Sheets (TDS) & Specification Pages

QC Category	Factor / Requirement	Impact Mechanism	Final QC Check	Y/N
Crawlability	Plain HTML text rendering (No JS-hidden specs)	Citations: AI crawlers cannot parse interactive sliders or collapsed JS accordions.	All specs visible in raw HTML page source.	
Fact-Hardening	Exact numerical bounds (Temp, Attenuation, Pitching Rate)	Citations: LLMs bypass vague claims ("fast yeast") in favor of specific numeric data.	Zero qualitative adjectives; all ranges have explicit units.	
Authority	Scientific method citation (Siebel, ASBC, EBC)	Citations: Strengthens source consensus score during retrieval-augmented generation (RAG).	At least 1 accredited R&D or analytical standard cited.	
Schema	JSON-LD Product & TechArticle implementation	Citations & Clicks: Feeds knowledge graph entities; enables Google Rich Results.	Schema passes Schema.org Validator without errors.	
Conversion	Above-the-fold CTA ("Calculate Dosing" / "Request Sample")	Clicks: Captures high-intent AI referral traffic landing directly on spec pages.	Primary CTA button positioned above the scroll line.	

2. Blog Posts & Informational Articles

QC Category	Factor / Requirement	Impact Mechanism	Final QC Check	Y/N
Structure	Sub-query H2 header mapping (Query Fan-Out)	Citations: AI breaks complex user prompts into sub-queries; H2s act as retrieval targets.	Every H2 matches a specific buyer sub-query.	
Summarization	3-bullet "Key Takeaways" box under H1 and H2s	Citations: Provides direct extractable blocks for instant citation engines.	Key Takeaway boxes present at top of each section.	
High-Yield Format	Inclusion of "Vs", "Best", "How-To", or comparison tables	Citations & Clicks: Formatted structures receive significantly higher AI citation volume.	Markdown table or structured list used for comparisons.	
Schema	JSON-LD FAQPage & Article schema	Citations & Clicks: Provides pre-parsed question-and-answer pairs for ai Summary	Validated FAQPage schema code present in HTML.	

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Freshness	Updated timestamp & verified parameter date	Citations: AI engines favor recency; older un-updated text suffers citation decay.	Published/updated date current within the last 90 days.	

3. Commercial Product Pages & Category Hubs

QC Category	Factor / Requirement	Impact Mechanism	Final QC Check	Y/N
SEO Metadata	Unique Title (< 60 chars) & Meta Description (< 155 chars)	Clicks: Drives organic SERP click-through rates (CTR) on traditional search engines.	Title contains target keyword; Meta Description has clear CTA.	
Off-Domain Signals	Unlinked branded mentions on Reddit, YouTube, forums	Citations: Strongest correlation factor (0.664) for inclusion in AI Overviews.	Verified third-party mentions on target platforms (r/TheBrewery).	
Entity Consistency	Matching product name, dosage, and brand entity across sites	Citations: Prevents AI confusion/hallucination across cross-engine summaries.	Name and specs match distributor catalogs and external TDSs.	
Bot Access	Unblocked robots.txt for AI crawlers (GPTBot, Google-Extended)	Citations: Prerequisite for AI indexing; blocked sites cannot be cited.	robots.txt explicitly allows target search and AI crawlers.	